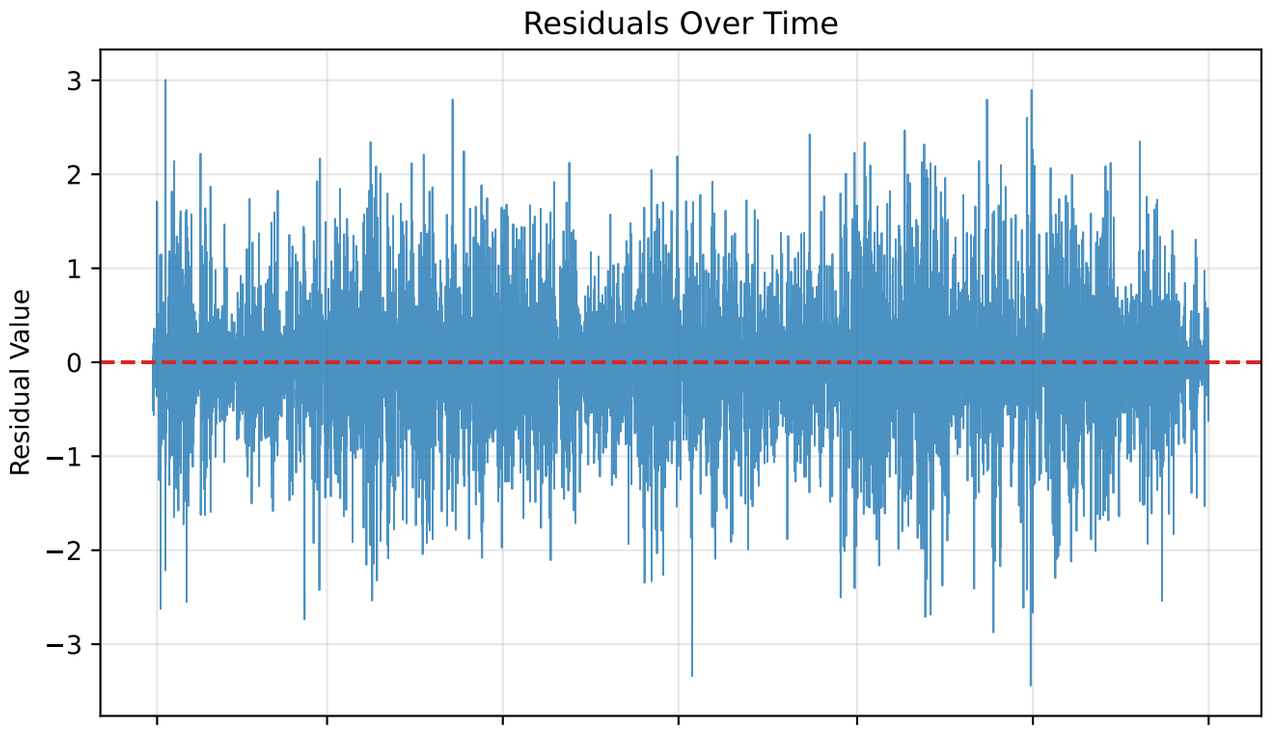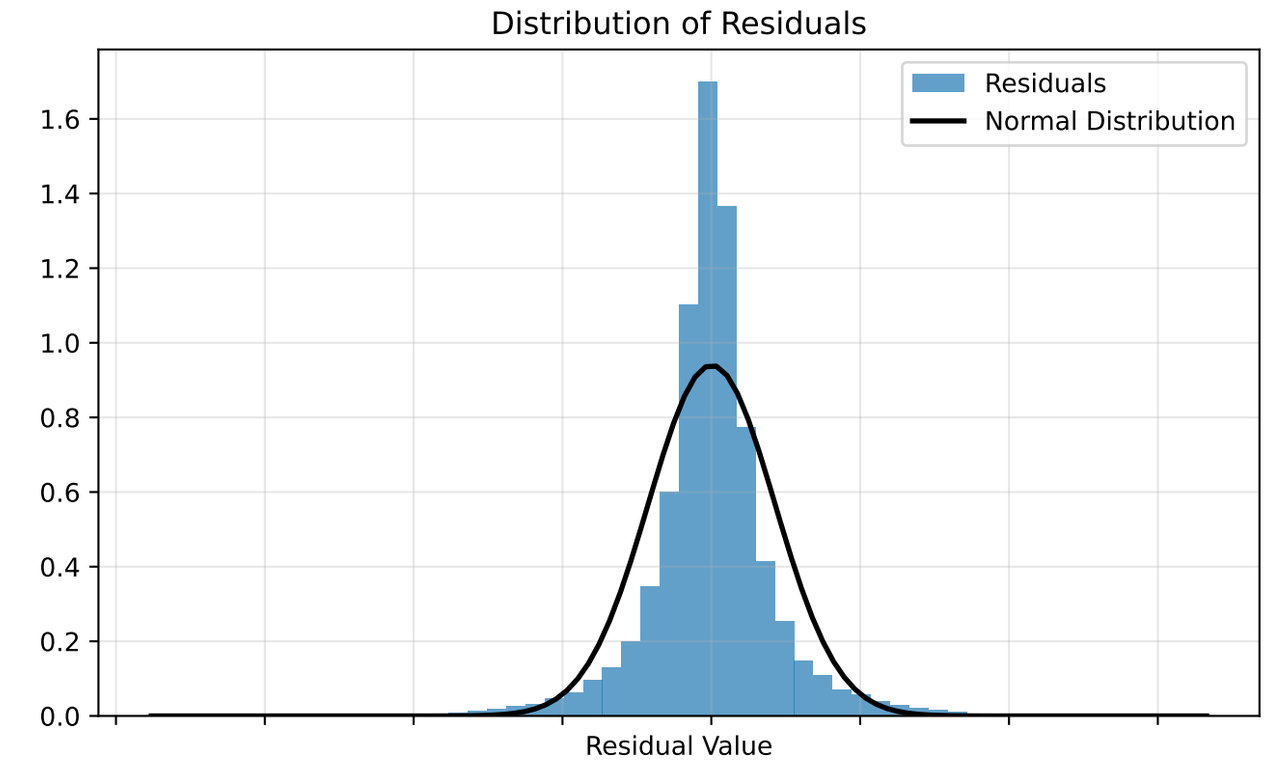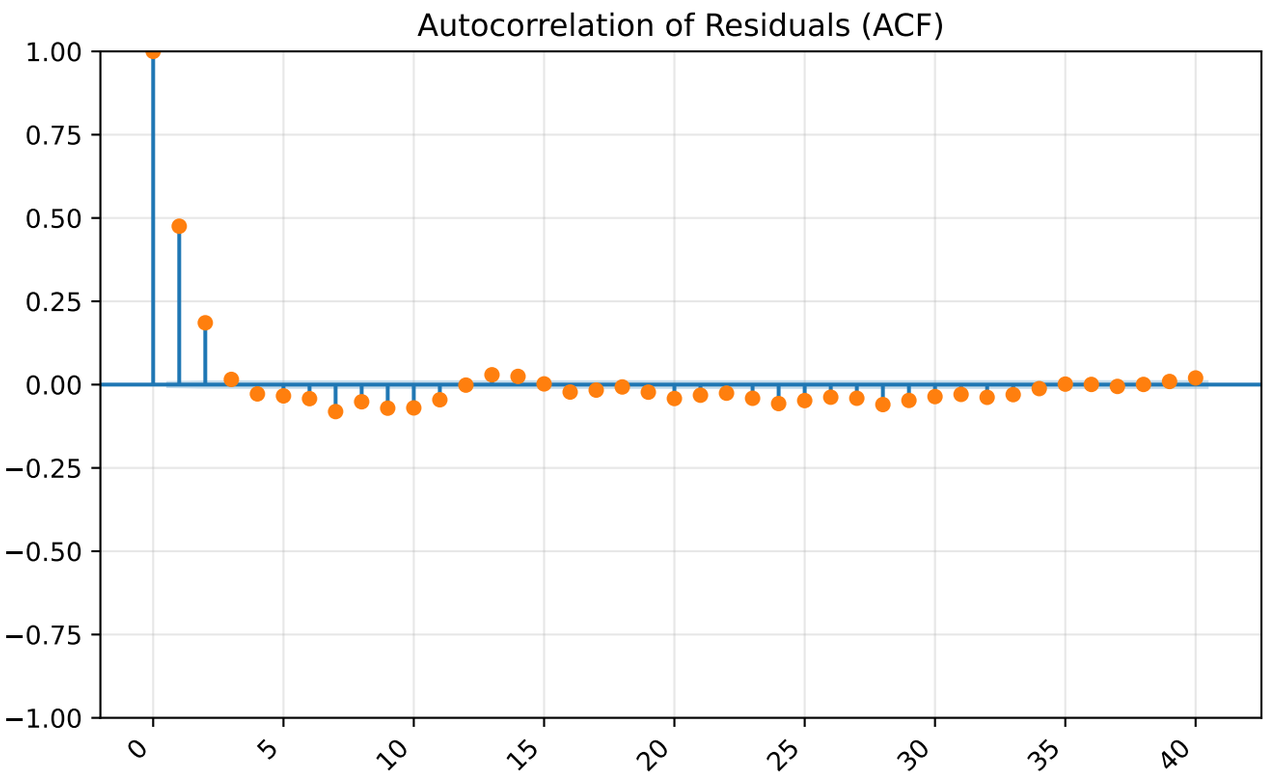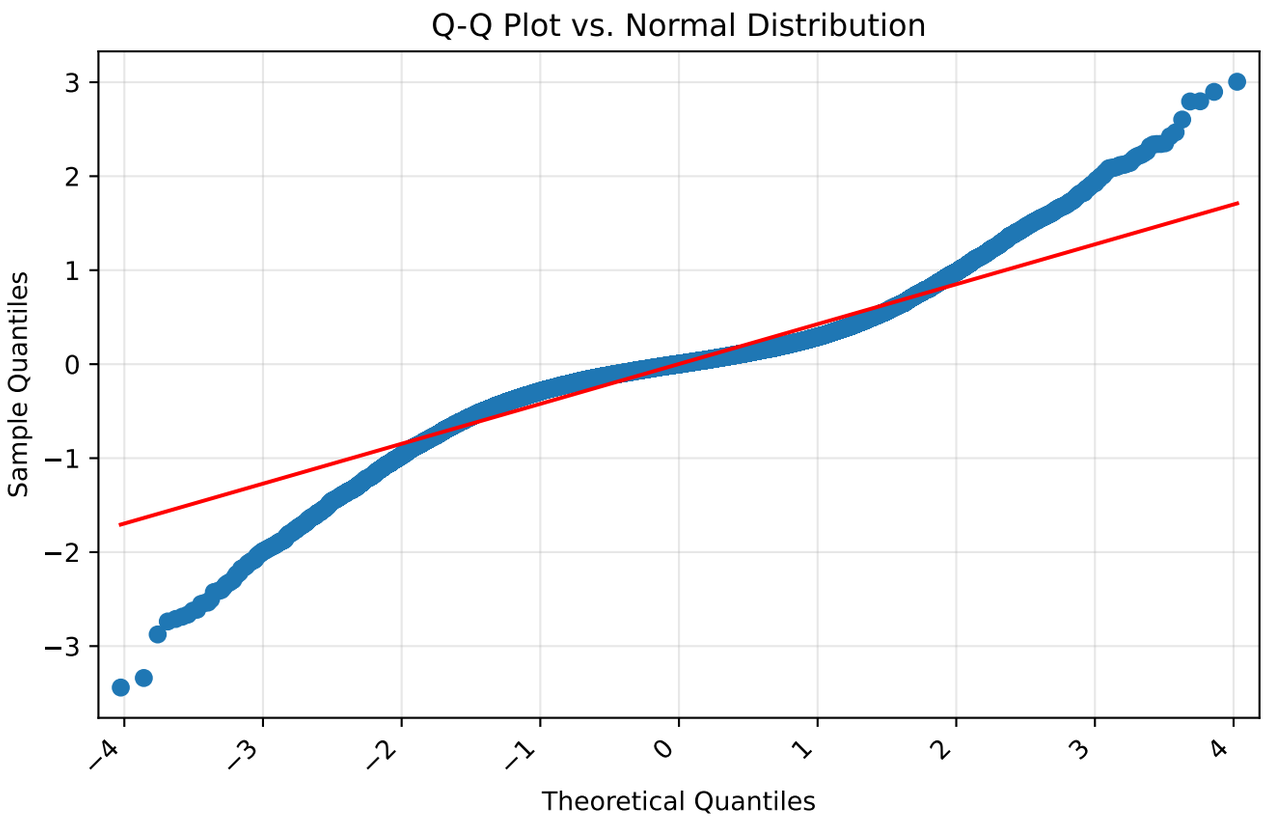Residual Diagnostics for 50Hertz (2023)